



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)



CSA1507 – CLOUD COMPUTING AND BIG DATA ANALYTICS

Quantum Computing as a Service

IBM (QCaaS): 

The Next Frontier of Cloud Platforms

Guided By
Dr.Gnana Soundari
SIMATS Engineering

Presented By
192525280 – P.Dharani Govardhan
192525169 – Nagella Jeevan Sai
192525105 – Praveen Kumar

INTRODUCTION

- Quantum Computing uses qubits instead of traditional bits.
- Qubits use superposition and entanglement to process information.
- QCaaS provides quantum computing resources through the cloud.
- Users can access quantum hardware without owning or maintaining it.
- QCaaS makes quantum computing more accessible, scalable, and cost-effective.



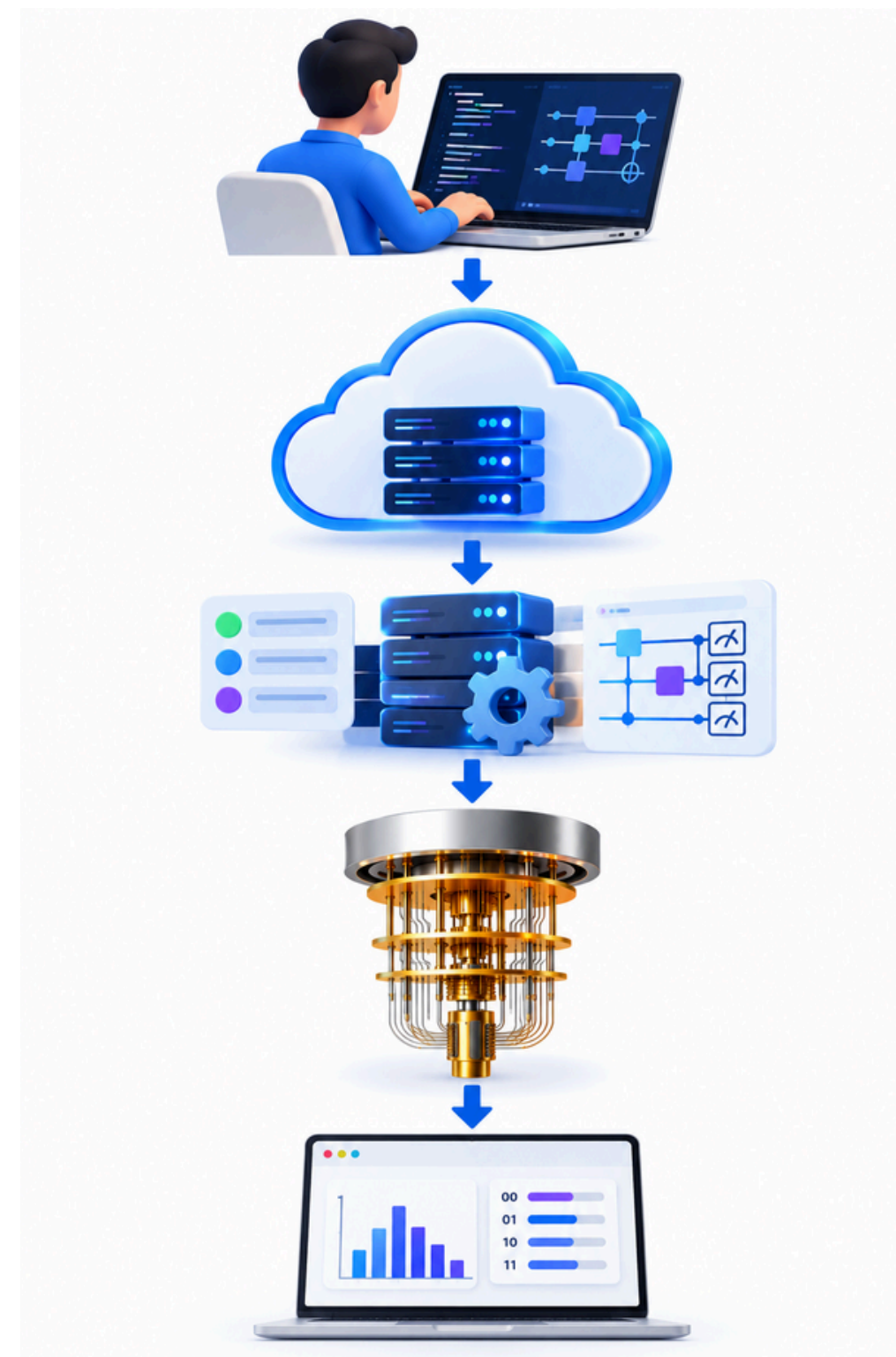
Why QCaaS?

- Expensive quantum hardware makes direct ownership difficult.
- QCaaS provides remote access to quantum computers through the cloud.
- Users can run quantum programs using cloud-based tools and APIs.
- It reduces the need for specialized infrastructure and maintenance.
- QCaaS makes quantum computing accessible to researchers, developers, and businesses.

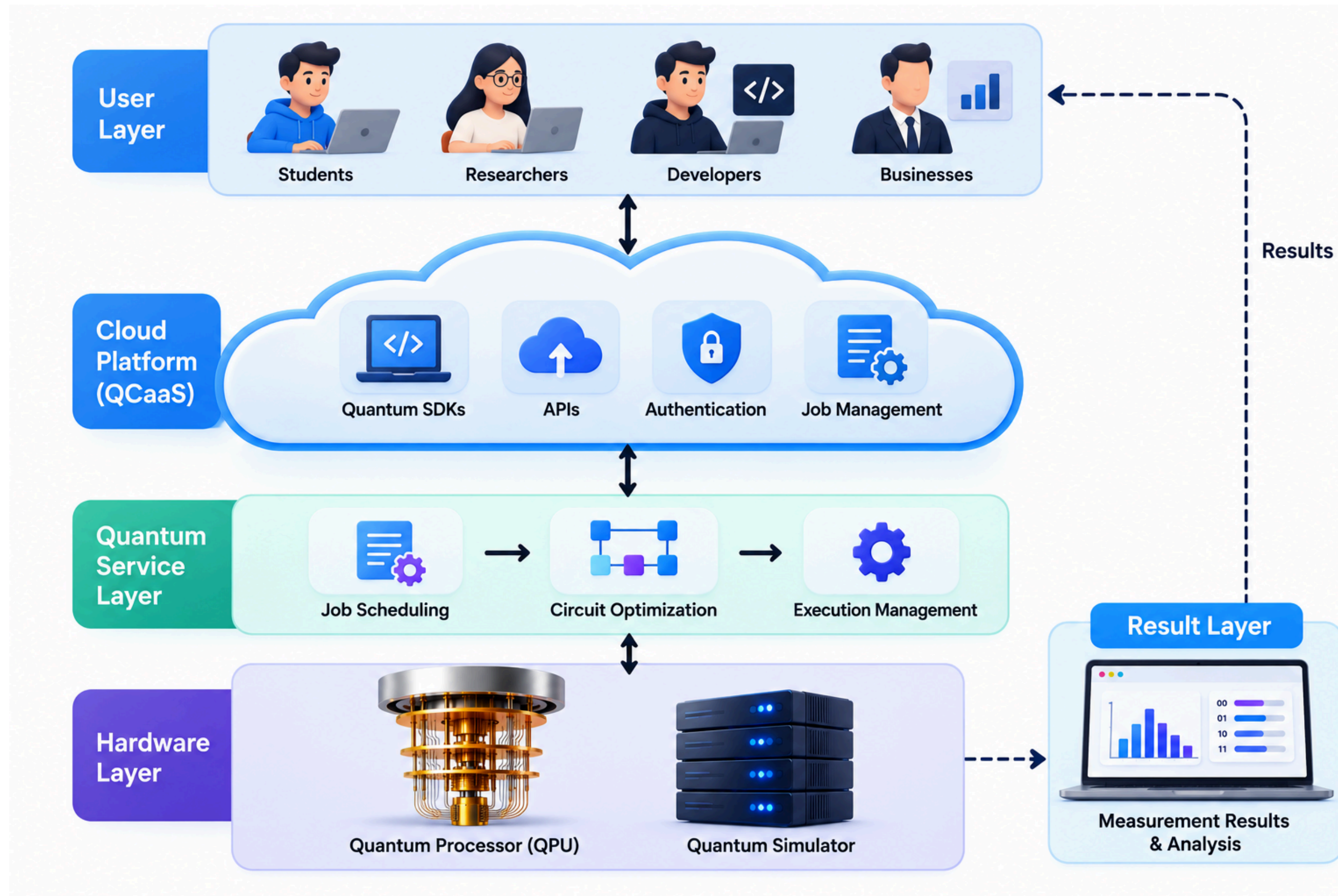


HOW QCAAS WORKS

- Quantum Program: User creates a quantum circuit using a quantum SDK.
- Cloud Submission: The program is sent to the QCaaS platform through the internet.
- Job Processing: The cloud platform schedules the job on a simulator or QPU.
- Quantum Execution: The quantum processor executes the circuit and measures the qubits.
- Result: Measurement results are returned to the user for analysis.



QCAAS ARCHITECTURE



BENEFITS OF QCAAS

- **Accessibility:** Access quantum computers through the cloud from anywhere.
- **Cost Efficiency:** Avoid the high cost of owning quantum hardware.
- **Scalability:** Use different quantum processors and simulators as needed.
- **Easy Integration:** Connect quantum computing with existing cloud and classical systems.
- **Faster Innovation:** Enables researchers and businesses to experiment with quantum technology easily.



REAL-WORLD APPLICATIONS OF QCAAS

- Drug Discovery: Simulating molecules and accelerating new drug research.
- Finance: Portfolio optimization, risk analysis, and financial modeling.
- Automotive: Developing advanced materials and improving EV battery research.
- Logistics: Optimizing routes, scheduling, and supply-chain operations.
- Healthcare & AI: Exploring complex simulations and quantum-enhanced algorithms.



REAL-LIFE EXAMPLE: IBM QUANTUM

- IBM Quantum provides real quantum computers through a cloud-based platform.
- Users can create quantum circuits using Qiskit and submit them through the cloud.
- Jobs can run on IBM quantum processors or simulators.
- Researchers and businesses can experiment without owning quantum hardware.
- IBM Quantum demonstrates how QCaaS brings quantum computing to real-world users.



LIMITATIONS OF QCAAS

- High Cost: Advanced quantum hardware and cloud access can still be expensive.
- Quantum Errors: Qubits are sensitive to noise and environmental disturbances.
- Limited Scalability: Current quantum processors have limitations in qubit count and reliability.
- Specialized Skills: Quantum programming requires new technical knowledge.
- Limited Use Cases: Quantum computing is not faster for every type of problem.



CONCLUSION

- QCaaS brings quantum computing to the cloud, making advanced quantum resources accessible remotely.
- It reduces the need for organizations to own and maintain quantum hardware.
- Real-world applications are emerging in automotive, finance, healthcare, chemistry, and logistics.
- Future improvements in qubit quality, error correction, and scalability can expand its practical use.
- QCaaS has the potential to become an important part of next-generation cloud platforms.



